“chaos” in the system) is used to control the range of movement for vertices on update, is computed for each vertex after an update, and it’s scaled down to smooth vertices oscillation.

Since GEM algorithm is not directly appliable as an instance of simulated annealing, we’ll stick with the algorithm developed by Davidson and Harel, which produces results of comparable quality.

As we mentioned in the past chapters, the first step with graph drawing algorithms is stating the criteria that will be used to judge the quality of an embedding: the crossing number is not the only key to draw graphs nicely. Davidson and Harel’s approach uses five criteria:

1. Distributing nodes evenly: spread the vertices uniformly in the canvas;
2. Keeping vertices away from borders;
3. Making edge-lengths uniform;
4. Minimizing edge-crossings;
5. Avoiding vertex-edges overlap by keeping vertices from coming too close to edges.

We must also clarify that the algorithm assumes the edges will be drawn as straight-line segments. Now, let’s see how these five criteria translate to formulas, by writing 5 components of the cost function.

For the first component, the algorithm uses a formula derived from electric potential energy; given two vertices v_i and v_j, we compute